

6th Grade Unit 11 Assessment Guide

General Information	The benchmarks on this assessment have assessment limits that may be helpful for the teacher to understand. For example, in the test item specifications, MA.6.AR.1.2 has an assessment limit that the given context should be continuous. This means that the contexts should not be true for just certain values as seen in the example in the benchmark, "Mrs. Anna told her class that they will get a pizza if the class has an average of at least 83 out of 100 correct questions on the semester exam. The inequality $g \geq 83$ can be used to represent the situation where students receive a pizza and the inequality $g < 83$ can be used to represent the situation where students do not receive a pizza." This context is not continuous because, as described, each question is correct or incorrect so only whole numbers are possible. This example would not be a type of problem students would see on the statewide assessment. The assessment limits for MA.6.AR.1.4 are "Items must include an expression with only one or two variables where coefficients, factors, and terms are integers only. Items will require the student to use one, all, or a combination of the associative property, distributive property, commutative property, or arithmetic operations to generate equivalent expressions." Teachers should note that the benchmark nor the assessment limits state that students should know when a property has been applied but that the items may require that the student employ one or more of the properties. Therefore, questions should not ask for a property name. The difficulty of the questions for MA.6.AR.1.4 are limited by the first assessment limit which limits the expression to one or two variables and all coefficients, factors, and terms are integers only. In grade 5, students evaluated multi-step numerical expressions using order of operations, MA.5.AR.2.2, where expressions were limited to no more than three operations that involve combinations of the four arithmetic operations and parentheses with whole numbers, decimals and fractions but could not include exponents or nested grouping symbols. Teachers should note that grade 6 students would be expected to answer questions for MA.6.AR.1.3 that go beyond what a 5th grader is expected to be able to do. This can be achieved by having questions that include integers, exponents, and/or nested grouping symbols. The questions on this assessment for MA.6.AR.1.4 includes at least one of these.
Calculator Information	All questions are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.6.AR.1.3
	☐ -22 ☐ -18 ☐ $(-2) - 20$ ☐ $(-2) + -20$ ☐ $2(-1) + 4(5)$ ☐ $2(-1) - 4(5)$	Recommended Scoring (12 Total Points)	Consider giving students 3 points for each correct choice. Consider deduction of all points for students who select all since the answers of -22 and -18 both could not be true.
Question	2	Benchmark(s)	MA.6.AR.1.3
	3	Recommended Scoring (6 Total Points)	Consider giving students 3 points for correct substitution. Watch for the common error of writing 2^3 as 6.
Question	3	Benchmark(s)	MA.6.AR.1.4
	☐ $2x + 18y$ ☐ $18y - 6x$ ☐ $18y - x + 3x$ ☐ $9(2y - x) + 3x$ ☐ $-2(3x - 8y) - 2y$	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct choice. Consider deduction of 2 points for each incorrect selection.

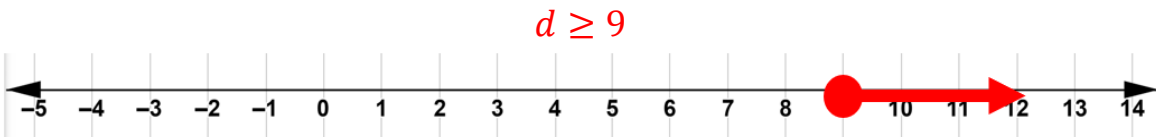
Question	4	Benchmark(s)	MA.6.AR.1.3
	5	Recommended Scoring (6 Total Points)	Consider giving students 2 points for correct substitution. Follow student work to determine if a minor error is carried throughout. Consider deducting 1 point for a minor error (such as losing a negative sign) that is carried out throughout. Monitor a student's application of the order of operations. Incorrect application of the order of operations should constitute a larger loss of points.

Question	5	Benchmark(s)	MA.6.AR.1.3
	-4	Recommended Scoring (6 Total Points)	Consider giving students 3 points for correct substitution. Watch for the common error of writing 10^3 as 30.
Question	6	Benchmark(s)	MA.6.AR.1.4
	☐ $2h + 33k$ ☐ $14h + 33k$ ☐ $8(3k + h) + 3(3k - 2h)$ ☐ $(8h - 6h) + (9k + 24k)$ ☐ $3(8k + 3k) - 2(-2h)$	Recommended Scoring (9 Total Points)	Consider giving students 3 points for each correct choice. Consider deduction of 2 points for each incorrect selection. Consider deduction of all points for students who select all since $2h + 33k$ and $14h + 33k$ as both could not be true.

Question	7	Benchmark(s)	MA.6.AR.1.3
	-78	Recommended Scoring (6 Total Points)	Consider giving students 3 points for correct substitution. Consider deducting 1 point for a minor error (such as losing a negative sign) that is carried out throughout. Monitor a student's application of the order of operations. Incorrect application of the order of operations should constitute a larger loss of points.

Question	8	Benchmark(s)	MA.6.AR.1.3
	191	Recommended Scoring (6 Total Points)	Consider giving students 3 points for correct substitution. Consider deducting 1 point for a minor error (such as losing a negative sign) that is carried out throughout. Monitor a student's application of the order of operations. Incorrect application of the order of operations should constitute a larger loss of points.

Question	9	Benchmark(s)	MA.6.AR.2.1
A. -7		Recommended Scoring (3 Total Points)	Consider not giving any partial credit.
Question	10	Benchmark(s)	MA.6.AR.2.1
☐ -4 ☐ -2 ☐ -1 ☐ 0 ☐ 2 ☐ 5		Recommended Scoring (12 Total Points)	Consider giving students 4 points for each correct choice. Consider deduction of 4 points for each incorrect selection.

Question	11	Benchmark(s)	MA.6.AR.1.2										
		Recommended Scoring (8 Total Points)	Consider giving students 4 points for the correct inequality. Consider deduction of 4 points for the correct graph.										
Question	12	Benchmark(s)	MA.6.AR.1.2										
		Recommended Scoring (8 Total Points)	Consider giving students 4 points for the correct inequality. Consider deduction of 4 points for the correct graph.										
Question	13	Benchmark(s)	MA.6.AR.1.2										
<table border="1" data-bbox="178 1036 1249 1253"><tbody><tr><td>$\frac{1200}{12}$ m</td><td>76.3 m</td><td>76.2 m</td><td>$74\frac{5}{9}$ m</td><td>76.05 m</td></tr><tr><td>76.5 m</td><td>$76\frac{1}{8}$ m</td><td>$\frac{7620}{1000}$ m</td><td>$76\frac{12}{16}$ m</td><td>$\frac{1500}{20}$ m</td></tr></tbody></table>		$\frac{1200}{12}$ m	76.3 m	76.2 m	$74\frac{5}{9}$ m	76.05 m	76.5 m	$76\frac{1}{8}$ m	$\frac{7620}{1000}$ m	$76\frac{12}{16}$ m	$\frac{1500}{20}$ m	Recommended Scoring (12 Total Points)	Consider giving students 2 points for the correct value. Consider deduction of 1 point for each incorrect value.
$\frac{1200}{12}$ m	76.3 m	76.2 m	$74\frac{5}{9}$ m	76.05 m									
76.5 m	$76\frac{1}{8}$ m	$\frac{7620}{1000}$ m	$76\frac{12}{16}$ m	$\frac{1500}{20}$ m									